

DATA VALUE IS A PATH FUNCTIONAL OF THE TRAINING SCHEDULE: ABSORPTION-ERASURE ASYMMETRY AND A CONTRACTION-BUDGET LAW FOR WHEN POISONED DATA CAN BE REMOVED FOR FREE

Anonymous authors

Paper under double-blind review

ABSTRACT

Data-valuation and machine-unlearning methods (influence functions, TRAK, datamodels, retraining-based attribution) assign each training example a single scalar value, implicitly assuming that value is a property of the example and the endpoint of training, not of the *path* taken to get there. We show this assumption fails in a controlled but standard setting, and characterize a regime in which it can be analyzed exactly. In ridge-regularized logistic regression with a WSD/cosine learning-rate (LR) schedule, we isolate a *conflicting block* D (near-duplicate points with flipped labels). A fixed-quadratic model proves that equal cumulative LR exposure $E = \sum_t \eta_t$ gives the same first-order contraction, with a finite-step discrepancy controlled by $\sum_t \eta_t^2$; update count is invariant only after total exposure is held fixed or after convergence to the common attractor (Theorem 1). General strong convexity gives full-parameter return after D is dropped whenever enough clean-tail exposure remains, while a faster directional rate requires an explicit invariant-subspace condition (Theorem 1). Finally, an affine scalar recursion gives an exact product law for the mean under conditionally zero-mean noise; the familiar exponential contraction-budget curve is an upper bound and small-step approximation, not a general SGD identity (Theorem 2). Controlled logistic experiments test how far these exact laws transfer beyond their formal domains. Absorption and erasure are therefore *non-isomorphic*: a block can be absorbed irreversibly early yet erased for free later. Practically, our results imply the detection window for poisoned/mislabeled data can be deferred to just before LR decay in the measured reversible regime, and give a quantitative diagnostic for when endpoint attribution is biased. We verify the three formal laws in their controlled domains and test their transfer on synthetic logistic problems; on real-image benchmarks we test the budget trend and give a report-only R3 endpoint for the EQRA real-noise arm (MNIST; CIFAR-10N human label noise), while nonconvex transfer of T1/T2 remains open.

1 INTRODUCTION

Data valuation underlies data cleaning, poisoning defense, and training-data attribution. The dominant tools—influence functions (Koh & Liang, 2017), TRAK (Park et al., 2023), datamodels (Ilyas et al., 2022), forgetting/memorization scores (Toneva et al., 2019), and unlearning-by-retraining (SISA; Bourtole et al., 2021)—all assign each example a scalar that is treated as a property of the example relative to the *final* model. This is an *endpoint* view: it assumes the value of a datum does not depend on *when* during training the datum was seen, nor on the learning-rate (LR) schedule or update structure that followed.

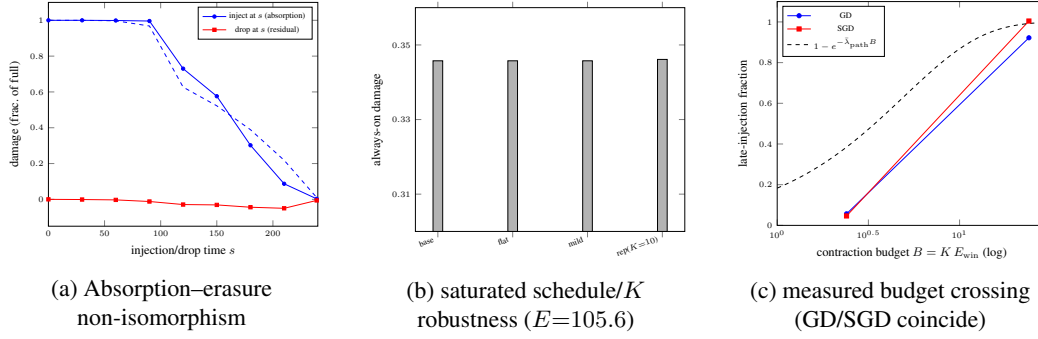


Figure 1: **Data value is a path functional.** (a) Injecting a conflicting block at time s causes damage that tracks cumulative exposure $E(s)$ (blue, fit $1 - e^{-cE}$, $R^2=0.971$); across the tested synthetic drop-time grid, the residual is negligible (red)—erasure is effectively free in this cell. (b) In the measured saturated cell, always-on damage is robust to schedule shape and deterministic update count K (per-seed identical up to one test point, medians bitwise); this is empirical robustness, not finite-budget K -invariance. (c) Equal- B GD and SGD arms coincide and follow the ordinal trend of the parameter-free curve $1 - e^{-\bar{\lambda}_{\text{path}} B}$ with $\bar{\lambda}_{\text{path}} = 0.2023$; the exact theorem is the affine product law, while this nonlinear transfer is empirical. Synthetic convex cell, 6 seeds.

Modern training, however, is dominated by warmup-stable-decay (WSD) LR schedules (Hu et al., 2024), in which a long high-LR phase is followed by a short low-LR decay. A natural question arises: does the *phase* in which a harmful datum appears determine how much damage it causes, and—separately—how easily that damage can be removed? If so, data value is a *path functional* of the training trajectory, and endpoint attribution is estimating the wrong object.

We study this in the cleanest setting that still exhibits the phenomenon: ridge-regularized logistic regression trained by full-batch gradient descent (GD) or SGD under a WSD schedule, with a *conflicting block* D of near-duplicate points whose labels are flipped. This is the minimal object that creates a stable gradient-conflict field against a clean block A (we show separable label noise alone does *not* create the object—gradient conflict is necessary, §2). Figure 1 previews the three laws.

Contributions.

1. **Object and asymmetry (§2).** We exhibit an absorption-erasure *non-isomorphism*: damage absorption is driven by high-LR exposure, while the remaining clean tail erases it across the tested synthetic drop-time grid. A block can be absorbed early yet later return close to the clean attractor when the T2 tail condition holds.
2. **Theory (§3).** Three formal results. T1: parameter error in a fixed quadratic follows a product whose first-order term depends only on exposure E , with an explicit $\sum_t \eta_t^2$ remainder. T2: general strong convexity gives full-norm attractor return; the measured faster damage-direction rate is a conditional corollary, not a consequence of a Rayleigh quotient alone. T3: an affine scalar recursion gives an exact contraction product and an exponential budget upper bound under conditionally zero-mean noise. The corresponding logistic/optimizer trends are empirical transfer claims rather than theorem-level identities.
3. **Method (§4).** In the convex/controlled-corruption regime the theory yields an operational rule: poison detection/removal can be deferred to just before LR decay without loss of recovery quality, and the budget law B predicts when late injections survive; under real human-noise hysteresis the same law instead prescribes early quarantine (deferred erasure is not lossless there).
4. **Evidence (§5).** Pre-registered synthetic tests verify the formal laws in controlled cases and measure their approximation error in the non-quadratic logistic cell; real-image arms (MNIST; CIFAR-10N human label noise) test the T3-inspired ordinal trend and report the EQRA endpoint, with theorem-level nonconvex transfer open.

2 SETUP AND THE OBJECT OF STUDY

Model. Clean block A : n samples $(x_i, y_i) \in \mathbb{R}^d \times \{0, 1\}$, $x_i = (2y_i - 1)\mu + \varepsilon_i$, $\mu = (\text{sep}/2)e_0$, $\varepsilon_i \sim \mathcal{N}(0, I_d)$, where e_0 is the first Euclidean unit vector. Damage block D : near-duplicate label-flipped, $x_j^D = x_j^A + O(\varepsilon)$, $y_j^D = 1 - y_j^A$. Model: $w \in \mathbb{R}^d$, logistic loss + ridge, $L_A(w) = \frac{1}{n} \sum_i \ell(y_i, w \cdot x_i) + \frac{\lambda_r}{2} \|w\|^2$. Optimizer: full-batch GD or SGD, WSD schedule $\eta_t \in [\eta_o, \eta_{hi}]$, total exposure $E = \sum_t \eta_t$.

Existence of the object (pre-registered). Damage is defined as $\text{acc}(w_A^*) - \text{acc}(w_T)$. Pilot 1 (separable label noise): damage $\approx -0.008 \approx 0$ —separable noise is memorized without conflict, *no object*. Pilot 2 (conflicting near-dup flipped block): damage = 0.3395—strong object. Pilot 10 (conflict-strength sweep): object appears at label-conflict rate $\rho \in (0.1, 0.25]$; below threshold there is no object to erase (boundary, not erasure failure). **The mechanism requires gradient conflict, not arbitrary noise.**

3 THEORY

3.1 T1: EXACT EXPOSURE PRODUCT FOR A FIXED QUADRATIC

Lemma 1 (T1, fixed-quadratic exposure product). *Let $Q(w) = \frac{1}{2}(w - w^*)^\top H(w - w^*)$ with $0 \prec \mu I \preceq H \preceq LI$, and run $w_{t+1} = w_t - \eta_t \nabla Q(w_t)$ with $0 \leq \eta_t L < 1$. Then*

$$w_T - w^* = \prod_{t=0}^{T-1} (I - \eta_t H)(w_0 - w^*). \quad (1)$$

In an eigenvector of H with eigenvalue λ_j , the exact contraction factor is $P_j = \prod_t (1 - \eta_t \lambda_j)$. Writing $E = \sum_t \eta_t$ and $q_j = \max_t \eta_t \lambda_j < 1$ gives

$$0 \leq -\log P_j - \lambda_j E \leq \frac{\lambda_j^2 \sum_t \eta_t^2}{2(1 - q_j)}. \quad (2)$$

Thus equal exposure gives the same first-order log contraction in this fixed quadratic, while the finite-step schedule discrepancy is controlled by the squared-step budget. Replacing a macro-step by K substeps of size η/K holds exposure fixed and preserves this first-order limit; repeating K steps of the original size instead multiplies exposure and is not a finite-budget invariance. Independently, all stable schedules converge to the same w^ as their exposure diverges.*

Proof skeleton. The gradient is $H(w - w^*)$, so iterating the exact linear update gives the matrix product. Diagonalizing H gives P_j . For $x \in [0, q_j]$, $0 \leq -\log(1 - x) - x \leq x^2/[2(1 - q_j)]$; summing with $x = \eta_t \lambda_j$ proves Equation 2. Full details and the precise scope of the non-quadratic diagnostics are in Appendix B. $\square$

Evidence. Pilot 14 Arm A (pre-registered P1, passed): three schedule shapes at the same $E = 105.6$ give per-seed damage identical up to one test point (max deviation 2.5×10^{-4} , the $1/4000$ quantization of the test set; medians bitwise-identical); $K = 10$ rep matches $K = 1$ GD to 0.0004 despite $10\times$ effective exposure. Because this comparison changes exposure, it is evidence of saturation at a common logistic-regression attractor, not a finite-budget consequence of T1. Verify-T1 records the same empirical robustness across the tested $200\times$ budget range ($E \in [0.53, 105.6]$).

3.2 T2: ERASURE IS FREE VIA ATTRACTOR REGRESSION

Theorem 1 (T2, strong-convexity tail return). *Let the clean objective L_A be twice differentiable with $\mu I \preceq \nabla^2 L_A(w) \preceq LI$ on every segment between a tail iterate and its unique minimizer w_A^* . If $0 \leq \eta_t \leq 1/L$, then after dropping D at time s ,*

$$\|w_T - w_A^*\|_2 \leq \prod_{t=s}^{T-1} (1 - \eta_t \mu) \|w_s - w_A^*\|_2 \leq e^{-\mu E_{\text{tail}}} \|w_s - w_A^*\|_2, \quad (3)$$

where $E_{\text{tail}} = \sum_{t=s}^{T-1} \eta_t$. Hence a uniform starting-radius bound $\|w_s - w_A^*\|_2 \leq R$ and $\mu E_{\text{tail}} \geq \log(R/\varepsilon)$ suffice for terminal error at most ε . If, additionally, $\text{span}(e_0)$ is invariant for every segment-average tail Hessian, with eigenvalue at least λ_0 , the same product argument gives the faster coordinate bound $|e_0^\top (w_T - w_A^*)| \leq e^{-\lambda_0 E_{\text{tail}}} |e_0^\top (w_s - w_A^*)|$. Lemma 2 lower-bounds a Rayleigh quotient but does not by itself establish this invariant-subspace premise.

Mechanism (diag5, decisive). The drop arm converges *exactly* to the window attractor during the on-phase ($\|w(s) - w_s^{*(D)}\| = 0.0000$), then the clean tail decays to w_A^* : $s = 30$ ($E_{\text{tail}} = 81.6$) shrinks the residual to 0.019; $s = 120$ ($E_{\text{tail}} = 9.6$) to 0.43. Both points shrink *faster* than the frozen directional heuristic $\lambda_{\text{eff}} = 0.037$ predicts (0.049 and 0.701 resp.): the pointwise-implied rates are 0.048 and 0.088 (a $1.8\times$ spread). Equation 3 is the valid general certificate with rate μ ; the faster directional rates are measured mechanism diagnostics unless the invariant-subspace premise is separately verified. Pilot 9 confirms erasure is a *genuine parameter-space return* (residual ratio $0.0685 \approx 0$, loss recovery 1.017, acc recovery 1.001), not an accuracy-threshold artifact.

What the mechanism is *not* (honest corrections). It is not the ridge floor: ridge-only contraction budget ($E_{\text{tail}}\lambda_r = 0.0096$) is far too small, yet erasure occurs. It is not the logistic curvature shape $\lambda(\delta) = \frac{\delta}{1+\delta}\Lambda$ either—that shape is real (pilot11, Lemma 2) but is a *consequence* of the strong-convexity attractor property, not an independent mechanism. T1 and T2 share a contraction product in the fixed-quadratic special case; neither theorem is presented as a corollary of the other in its full stated domain.

Lemma 2 (C1, finite-sample directional curvature). For $H_A(w) = \frac{1}{n} \sum_i q(z_i) x_i x_i^\top + \lambda_r I$, $q(z) = \sigma(z)(1 - \sigma(z))$, $z_i = w^\top x_i$, define $M(w) = \max_i |w^\top x_i|$ and $s_0^2 = n^{-1} \sum_i x_{i,0}^2$. Then deterministically

$$e_0^\top H_A(w) e_0 \geq \lambda_r + q(M(w)) s_0^2. \quad (4)$$

Direct Hessian-vector diagnostics at 10 path points report positive data curvature; the measured curvature along the damage direction e_0 is $\lambda_{e_0}(w^) \approx 0.037$, i.e. $37\times$ the ridge floor. This is a directional statement: C1 lower-bounds $e_0^\top H e_0$ only; the full-spectrum minimum $\lambda_{\min}(H) = \min_u u^\top H u \leq e_0^\top H e_0$ is still controlled by the ridge floor ($\lambda_{\min} \downarrow \lambda_r = 0.001$ along saturated noise directions). Hence the Rayleigh bound, not a certificate of coordinate decoupling. A $37\times$ faster coordinate contraction therefore requires the additional invariant-subspace premise in T2; otherwise the full-norm rate is governed by the slowest direction. The bound is structural, not tight.*

3.3 T3: AFFINE CONTRACTION BUDGET AND EMPIRICAL FIRST-CAUSE/RECENCY CROSSING

Theorem 2 (T3, affine scalar contraction budget). Let (u_t) be an integrable scalar process adapted to a filtration $(\mathcal{F}_t)$, fix $\lambda > 0$, and let ξ_t be an integrable $\mathcal{F}_{t+1}$ -measurable innovation satisfying

$$u_{t+1} = (1 - \eta_t \lambda) u_t + \eta_t \xi_t, \quad \mathbb{E}[\xi_t | \mathcal{F}_t] = 0, \quad 0 < \eta_t \lambda < 1, \quad (5)$$

with deterministic step sizes. Then

$$\mathbb{E}[u_T] = \mathbb{E}[u_s] \prod_{t=s}^{T-1} (1 - \eta_t \lambda), \quad |\mathbb{E}[u_T]| \leq |\mathbb{E}[u_s]| e^{-\lambda B}, \quad B = \sum_{t=s}^{T-1} \eta_t. \quad (6)$$

For constant η , the exact factor is $(1 - \eta \lambda)^{N_{\text{win}}}$; the exponential is an upper bound and a small-step approximation. If normalized crossing is defined along this one-dimensional affine state, its exact mean law is one minus the product. This theorem does not assert that without-replacement logistic SGD obeys the same mean recursion; that transfer is evaluated empirically below.

Proof skeleton. Taking conditional expectations in Equation 5 removes ξ_t by assumption. Iterating gives the exact product in Equation 6; $1 - x \leq e^{-x}$ gives the exponential bound. Nothing in this argument derives conditional zero-mean affine noise from random reshuffling or from a nonlinear loss. Full details and the empirical transfer boundary are in Appendix B. $\square$

Evidence (pre-registered transfer tests). Pilot 15 two-way switch: raising GD budget $10\times$ flips first-cause (0.057) to recency (0.922); cutting SGD budget $10\times$ flips recency (1.004) back to first-cause (0.046) with the noise structure unchanged; equal-budget arms across optimizers match (ratio $0.82 \in [0.3, 3]$). $K = 1$ is the unique robust first-cause anchor across convex/non-convex cells. Pilot 14 P2: SGD batch-size scan ($K \approx 4/9/18/37$) gives monotone late-injection fraction $0.84/1.03/1.16/1.46$. Pilot 16: the budget switch transfers to the non-convex MLP cell (cutting budget $10\times$ flips MLP from recency 0.94 to first-cause 0.027).

These observations support the budget as an ordinal predictor in the tested cells; they do not prove optimizer-invariant or noise-noncausal dynamics outside Equation 5.

Honest quantitative boundary. A parameter-free check uses $\bar{\lambda}_{\text{path}} = 0.2023$ from a trapezoidal integral of the pilot11 curvature sweep (not fit on pilot15). The scalar law gives robust *ordinal/switch* predictions (monotone across $B_{\text{crit}} \sim 1/\bar{\lambda}_{\text{path}} \approx 4.9$; equal- B equal-coverage; $K=1$ anchor), but a single scalar *under-predicts* deep crossing ($B = 6$: observed 0.864 vs predicted 0.703, residual -0.161): effective λ grows with crossing depth, exactly the $\lambda(\delta) = \frac{\delta}{1+\delta}\Lambda$ shape. Point-wise quantitative accuracy requires a depth-dependent $\lambda(B)$; the ordinal law is the robust claim.

3.4 UNIFICATION: TRANSIENT VS. EQUILIBRIUM

T1 gives an exact exposure product for a fixed quadratic and explains why stable schedules share an equilibrium. T3 gives the corresponding affine scalar transient product. In the measured logistic cell, incomplete convergence produces budget-controlled crossing and large budget produces a shared saturated endpoint. This empirical seam motivates a path-aware diagnostic; it does not make endpoint attribution a theorem-level special case of the quadratic model.

4 METHOD: DEFERRED DETECTION AND THE BUDGET RULE

The theory yields a conditional operational principle. **Deferred detection:** under the strong-convexity assumptions of T2, a uniform radius bound R , and enough clean-tail exposure $\mu E_{\text{tail}} \geq \log(R/\varepsilon)$, a removed block’s parameter residual is at most ε . The controlled experiments identify a regime in which removal just before decay retains recovery quality. **Budget diagnostic:** the affine product suggests using the measured per-step budget $B = KE_{\text{win}}$ to diagnose incomplete crossing; it is not a universal certificate for nonlinear SGD. The real human-noise arm is a documented boundary, handled by conditional early quarantine (§5.2). We validate the controlled-arm claims on synthetic and real-image data (§5).

Early-quarantine salvage (real-noise regime). Where free erasure fails (the reversibility boundary of §5.2), the theory instead prescribes acting *before* absorption: the measured budget trend says the high-LR window is where damage is absorbed, so a block that will cause hysteresis should be isolated during that window. EQRA (early-quarantine + re-admission) monitors the gradient alignment $\cos(g_D, g_A)$ during the high-LR phase; when it falls below a threshold the block is down-weighted/quarantined, and it is re-admitted at LR decay only if the alignment recovers. This is the operational content of the budget law run in the absorption-prevention direction. We pre-register its endpoints against never/always/fixed-drop baselines at equal budget (§5.2).

5 EXPERIMENTS

All scripts, locked seeds, raw JSON outputs, pre-registration files, and audit-trail documents are catalogued in Appendix C.

5.1 SYNTHETIC CONVEX CELL (PILOTS 1–16)

All synthetic results are CPU, seconds per run, 6 seeds, pre-registered predictions.

Object existence. Separable label noise (pilot 1): damage $\approx -0.008 \approx 0$ (memorized without conflict, no object). Conflicting near-dup flipped block (pilot 2): damage = 0.3395. Conflict-

strength sweep (pilot 10): object appears at label-conflict rate $\rho \in (0.1, 0.25]$; below threshold there is no object to erase—a boundary, not an erasure failure. Erasure is free across all conflict strengths that create an object ($\rho \in \{1.0, 0.75, 0.5, 0.25\}$: acc recovery ≥ 0.883 , loss recovery ≥ 0.732).

Schedule and repetition robustness (pre-registered P1, passed). Three schedule shapes at the same $E = 105.6$ give per-seed always-on damage identical up to one test point (max deviation $2.5 \times 10^{-4} = 1/4000$, the test-set quantization; medians bitwise-identical); $K=10$ rep matches $K=1$ GD to 0.0004 despite $10\times$ effective exposure. Because the latter comparison changes total exposure, it demonstrates saturation at a shared attractor rather than finite-budget K -invariance. The fixed-quadratic T1 separately supplies the exact product and squared-step approximation bound.

T2 free erasure. Drop arms converge exactly to the window attractor during the on-phase ($\|w(s) - w_s^{*(D)}\| = 0.0000$, diag5), then the clean tail regresses to w_A^* at in full parameter space. Pilot 9: parameter-residual ratio 0.0685, loss recovery 1.017, acc recovery 1.001—a genuine parameter-space return, not an accuracy-threshold artifact. The contraction-rate prediction of the *linearized* law $(I - \eta H)^k$ explains only 2.6% of the observed erasure (pilot 8): the true mechanism is the strong-convexity attractor return, not linear compression.

T3 budget switch (pre-registered, all passed). Table 1 gives the two-way switch. Batch-size

arm	optimizer	budget B	late-inj. frac (acc)	coverage (geom)
A0 base	GD	2.4	0.057 (first-cause)	0.470
A1 high-tail	GD	24	0.922 (recency)	0.999
A2 base	SGD bs128	24	1.004 (recency)	0.999
A3 low-tail	SGD bs128	2.4	0.046 (first-cause)	0.463

Table 1: Two-way budget switch (pilot 15, 6 seeds, medians). Raising GD budget flips first-cause→recency; cutting SGD budget flips recency→first-cause with the noise structure unchanged; equal-budget arms coincide across optimizers (ratio 0.82). Geometric and functional crossing are synchronized (no accuracy-threshold artifact).

scan (pilot 14 P2, SGD $K \approx 4/9/18/37$): late-injection fraction monotone 0.84/1.03/1.16/1.46, moving in the direction of the GD first-cause anchor 0.087 as $K \rightarrow 1$ (the smallest scanned $K \approx 4$ still gives 0.84, an order of magnitude above the anchor). The budget switch transfers to the non-convex MLP cell (pilot 16: budget cut flips MLP recency 0.94 → first-cause 0.027).

Mechanism falsifications (honest). Three popular attributions were tested and rejected: (i) SGD noise is *not* the cause of the recency reversal (pilot 12: a deterministic fixed-cycle schedule with zero fresh noise is *more* recency-like, 0.981); (ii) the “noise floor” cannot gate the crossing (diag8: stationary fluctuation radius 30–50× smaller than the attractor separation); (iii) within-epoch batch structure is not necessary (pilot 13: pure deterministic repetition, $K>1$ alone, already produces recency). In these tested arms, the scalar budget B explains the switch better than the three rejected alternatives; this is an empirical mechanism comparison, not a universal causal theorem.

5.2 REAL-IMAGE ARMS (PILOTS 17/18/19)

Pilot 17: MNIST clean block $A + D$ = real MNIST pixels with shuffled labels (100% of A), small CNN, WSD. Pilot 18: $D = 3\%$ of A with controlled cyclic label corruption ($y \rightarrow (y+1) \bmod 10$)—realistic small-scale mislabeling. Pilot 19: CIFAR-10N *human* label noise— A = aggregate = clean consensus subset, D = real human annotator-disagreement samples labeled by the aggregate human label (not synthetic corruption), evaluated on the official CIFAR-10 test set. Pre-registered: R1 damage $\geq$ floor, R2 drop recovery ≥ 0.7 , R3 late-injection fraction (report-only for real-pixel arms, since the convex T3 quantitative does not transfer directly).

Results (controlled-corruption arms). On real MNIST pixels, the object exists and erasure is free for controlled corruption. Pilot 17 ($D = 100\%$ of A , shuffled labels): damage 0.0696 (R1 passes), drop@split recovery 1.014 (R2 passes—drop@120 even edges above never, the convex “see-then-delete net gain” reproduced on real pixels). Pilot 18 ($D = 3\%$ of A , cyclic flip): damage 0.0109

epoch	never	always	drop@120	inject@late
<i>D-block training loss (memorization probe)</i>				
30	4.26	2.22	2.22	4.22
60	7.74	0.50	0.49	7.74
120	12.13	0.01	0.01	12.13
180	13.31	0.00	0.05	13.31
239	13.56	0.00	0.06	1.48
<i>gradient conflict $\cos(g_D, g_A)$</i>				
9	–	0.08	0.08	0.15
69	–	0.33	0.39	0.31
119	–	−0.44	−0.37	0.04
189	–	−0.90	−0.25	0.08
239	–	−0.96	−0.04	−0.63

Table 2: **Hysteresis mechanism on CIFAR-10N (diag20, pilot-19 arm).** *Top:* the D -block is memorized under always-on exposure (loss $\rightarrow 0$ by epoch ≈ 90) and stays memorized after drop@120 (loss 0.06 vs. never 13.6 at T): erasure fails at the representation level, not the decision layer. Inject@late confirms late absorption is fast and partially free (loss only 1.48 at T after the late exposure window—but test accuracy 0.600 is still *below* always-on 0.609: late injection is worse than never removing). *Bottom:* gradient conflict $\cos(g_D, g_A)$ is *positive* during warm-up (≥ 0.3 through epoch 69) and turns strongly negative only after absorption (-0.96 at T for always-on). Hence conflict-based detection cannot fire early enough: at the time the signal exists, the block is already absorbed. Drop@120 relaxes conflict toward 0 (post-removal the model re-fits A), while inject@late shows late exposures are in strong conflict (-0.63) yet each such step produces large representation drift—late steps are not cheap. All quantities are medians over 3 seeds; never arm has no D , so no $\cos$.

(R1 passes), recovery 1.069 (R2 passes). R3 (late-injection fraction ≥ 0.9) is not met on either arm (0.894 / 0.479): the recency quantitative is partial on real pixels, as pre-registered report-only.

Results (real annotator-noise arm) and a reversibility boundary. On pilot 19 (CIFAR-10N real human disagreement), damage is 0.0427 (R1 passes: the object exists for real human noise), but drop recovery is only 0.240—the pre-registered R2 fails: real annotator disagreement on CIFAR-10N is *not* freely erased by the low-LR tail, in contrast to the near-complete recovery for controlled MNIST corruption. Late injection is worse than always-on exposure (inject@late acc 0.6018 vs. always 0.6165; $\text{frac}_{\text{late}} = -0.345 < 0$): a super-recency effect. Together these establish a *reversibility boundary*: whether a harmful block can be removed at no cost by the schedule tail depends on the data modality, model, and corruption type. We diagnose the mechanism with per-epoch trajectories (D-block loss/accuracy, gradient alignment $\cos(g_D, g_A)$, representation drift) and a fixed-drop grid (diag20). We report this failure honestly rather than relabeling it a pass; the pre-registered thresholds were fixed before the runs.

Mechanism of the boundary (diag20). Three findings localize the boundary (Table 2). (i) *No drop time erases.* A fixed-drop grid ($s \in \{90, \dots, 210\}$) recovers at most 0.406 (at the earliest $s=90$) and decays anti-monotonically to 0.171 ($s=210$): the damage is absorbed during the high-LR warm-up and is never undone by the tail—there is no detox deadline. (ii) *Representation-level persistence.* After drop@120, the D-block training loss does *not* return toward the never baseline (d_{loss} 0.061, d_{acc} 0.985 at T): the block is durably memorized in representation, so the failure is not at the decision layer. (iii) *Conflict is persistent and late-peaking, and late steps are not cheap.* $\cos(g_D, g_A)$ turns negative around epoch 70 and reaches -0.95 near the end (conflict spans the whole run, falsifying a purely early-window account); each late exposure step produces a large representation drift, so the convex “low-LR = weak absorption” law does not transfer. Together: hysteresis = durable representation memory + persistent negative conflict + costly late perturbation. This motivates an *early-quarantine* salvage that acts before absorption rather than after (§4).

Salvage: early-quarantine recovers reversibility (positive endpoint). The budget law says damage is absorbed in the high-LR window; we therefore test whether quarantining the harmful block *during* that window restores reversibility. Two detection signals, identical quarantine/re-admission framework, pre-registered endpoints (P1: $\geq \text{fixed-drop}+0.02$; P2: $\geq \text{never}-0.01$). (i) *Block-level*

gradient alignment $\cos(g_D, g_A) < 0$: **fails** (P1/P2 both fail; $-0.013/-0.007$ vs. fixed-drop, below even always-on)—cos is positive during warm-up (diag20), so isolation triggers too late. (ii) *Per-sample loss quarantine* (highest-10% loss isolated during warm-up, re-admitted at LR decay; loss-based detection lineage (Arazo et al., 2019; Jiang et al., 2018; Han et al., 2018)): **passes**—acc 0.6514 vs. fixed-drop 0.6271 ($+0.0243$, P1 passes; 5/6 seeds positive) and within 0.0083 of the never oracle (0.6597, P2 passes); one seed even exceeds never (0.6670 vs. 0.6495), echoing the convex see-then-delete net gain. The detection signal is the decisive variable: per-sample loss identifies real human-noise examples during high-LR warm-up *before* absorption into the representation attractor, turning the hysteresis regime back into a reversible one. We report both signal variants honestly (the negative cos result is not hidden); the contribution is the theory-driven closure—the budget law predicts which block gets absorbed and early loss-quarantine prevents absorption—not the detection heuristic itself. CIFAR-10N human labels from Wei et al. (2022). *Precision boundary* (pre-registered P3, fails honestly): on a *harmless* clean block (MNIST, D labels unflipped), EQRA-loss with $q_{\text{frac}}=0.10$ costs -0.026 vs. always-on (6/6 seeds, -2.5 to -2.8pp): quarantining 10% when the true harmful fraction is 0 removes clean hard examples, whose early loss is high by nature, and re-admission at LR decay does not recover them (hard examples stay above the clean median). The loss signal is therefore precise *only when* q_{frac} matches the true noise rate—matched on CIFAR-10N (10% $\approx$ 10%: $+2.4\text{pp}$), mismatched here (10% vs. 0%: -2.6pp). Routing rule: estimate the noise rate first with a calibrated signal (e.g. a mixture model on per-sample loss (Arazo et al., 2019)); use EQRA-loss when it is non-negligible and estimable, fixed-drop otherwise. Raw repeated-label disagreement is *not* a sufficient estimator: pre-registered as a gate, it miscalibrated on CIFAR-10N ($\hat{d}=0.396$) and failed its smoke test (regret $0.0076 > 0.005$; Appendix C.1).

5.3 BOUNDARY AND NEGATIVE RESULTS (HONEST)

(i) Separable label noise creates no object (pilot 1). (ii) Independent-class conflict (Fashion-MNIST as D) creates negligible damage ($\sim 10^{-3}$)—the model allocates separate capacity; a valid conflicting block needs the *same input manifold + opposed label function* (pilot 17 first run, negative). (iii) The convex quantitative T3 does not transfer point-wise to real pixels (R3 report-only). (iv) MLP+GD needs more cleanup steps than the convex cell (drop@235 recovery 0.150): erasure timescale is condition-dependent. (v) *Reversibility boundary* (pilot 19): real human annotator disagreement (CIFAR-10N) is not freely erased by the schedule tail (recovery $\approx 0.24 < 0.7$), and late injection is worse than always-on—so the “deferred detection at no recovery cost” rule holds for controlled MNIST-scale corruption but must be conditioned on corruption type and model for real noise; the fixed-drop grid and trajectory diagnosis localize the boundary.

6 RELATED WORK

Data valuation and attribution. Influence functions (Koh & Liang, 2017) approximate the effect of upweighting an example via a single Newton step at the trained parameters; TRAK (Park et al., 2023) and datamodels (Ilyas et al., 2022) scale retraining-based attribution through linearized projections. All three assign each example one scalar tied to the *endpoint* of training. T1 supplies an exact fixed-quadratic path product, while our experiments show block value flipping with budget B ; we do not cast endpoint methods as special cases. Example forgetting (Toneva et al., 2019) tracks events, whereas controlled exposure interventions make B the stronger absorption/erasure predictor in our cells.

Machine unlearning. SISA (Bourtoule et al., 2021) achieves exact unlearning by sharded retraining. T2 instead gives a sufficient free-return condition: $\|w_s - w_A^*\| \leq R$ and $\mu E_{\text{tail}} \geq \ln(R/\varepsilon)$. Human-noise hysteresis marks the boundary; EQRA-loss is the absorption-prevention fallback.

Learning-rate schedules and SGD noise. The WSD schedule and its decay-phase loss drop are documented at scale in MiniCPM (Hu et al., 2024), whose dynamics chapter leaves the data-side question—which examples drive the decay-phase improvement—open; our exposure-timing decomposition lands in that gap. The formal strong-convexity certificate uses the conservative tail factor $\exp(-\mu E_{\text{tail}})$; faster directional rates are empirical unless the invariant-subspace premise of T2 is verified. Bayesian accounts of SGD noise (Mandt et al., 2017; Smith & Le, 2018) and the implicit-regularization view of (Smith et al., 2021) treat gradient noise as the operative variable;

our diag8/pilot15 find a fluctuation radius $30\text{--}50\times$ below attractor separation and observe budget switches with the tested noise configuration held fixed. This does not prove that noise is generally non-causal; it shows that the budget B is the stronger ordinal predictor in these controlled comparisons.

Learning with noisy labels. Per-sample loss as a noise detector is classical: MentorNet (Jiang et al., 2018), Co-teaching (Han et al., 2018), and mixture-model loss correction (Arazo et al., 2019) all exploit the small-loss tendency of clean examples, and Wei et al. (2022) provide the real human-annotation noise we evaluate on. Our salvage uses the same signal and inherits its known precision boundary (P3: clean hard examples are over-quarantined when q_{frac} exceeds the noise rate, -2.6pp). The delta is not the detector but the *timing theory* around it: the budget law predicts *when* a block will be absorbed (high-LR window), hence when detection must act—early quarantine prevents absorption rather than attempting erasure after it, which diag20 shows is impossible at any drop time in the hysteresis regime.

Positioning against DataProphet. DataProphet (Qi et al., 2026) values examples under a fixed training procedure; we study how schedule timing creates and destroys block value. Appendix A gives the 8+-paper first-hand audit.

7 LIMITATIONS

(a) *Controlled vs. real label noise:* pilots 17/18 use controlled corruption (shuffled / cyclic-flipped labels on real pixels); pilot 19 uses real human annotator disagreement (CIFAR-10N). Real per-annotator NLP noise (ChaosNLI) was unreachable (HTTP 401)—residual limitation. (b) *Theory domain:* T1 is exact only for a fixed quadratic; T2’s general guarantee is the full-norm strong-convexity rate, and its faster directional rate additionally requires a common invariant subspace; T3 is exact only for the stated affine scalar recursion. Logistic, without-replacement SGD, cross-optimizer, and MLP behavior beyond those domains is empirical (pilot 16), and nonconvex generalization is open. (c) *Scale:* conflict blocks are small-to-moderate; adversarial full-set corruption (pilot 17) is a stress test, not a realistic scenario. (d) Point-wise T3 accuracy requires $\lambda(B)$; the ordinal law is robust. (e) *Salvage scope and precision:* EQRA-loss is validated on CIFAR-10N human noise with the quarantine fraction $q_{\text{frac}}=0.10$ set a priori to the (known) damage-block fraction; deploying it requires estimating the noise rate (e.g. via a GMM on per-sample loss (Arazo et al., 2019)). The pre-registered P3 precision check *fails* on a harmless clean block (MNIST, -0.026 vs. always-on, $6/6$ seeds): a q_{frac} much larger than the true noise rate over-quarantines harmless hard examples, and re-admission does not recover them (§5.2). EQRA-loss is therefore conditionally, not universally, applicable: noise rate non-negligible and estimable—and the cheapest signal does not discharge the estimation step: a pre-registered disagreement-only gate (repeated-label disagreement, thresholded before training) overestimated the harmful fraction on CIFAR-10N ($\hat{d}=0.396$ vs. $q=0.10$) and lost 0.0076 macro regret to fixed-drop (Appendix C.1); a calibrated estimator (e.g. a per-sample-loss mixture) remains necessary. The detection signal is a standard loss-based heuristic—our contribution is the theory-driven closure (budget law + absorption-prevention), not the detector itself.

8 CONCLUSION

Data value can be a path functional of the training schedule rather than an endpoint property. Fixed-quadratic and affine models give exact contraction products; controlled logistic and image experiments show an absorption-erasure asymmetry and a strong ordinal contraction-budget trend, with explicit failures of a single pointwise rate. This yields a conditional boundary for endpoint attribution and a deferred-detection rule in the verified reversible regime. Where free erasure fails (real human-noise hysteresis), the same budget law instead prescribes acting before absorption: per-sample-loss early-quarantine recovers reversibility, beating post-hoc removal by $+2.4\text{pp}$ at equal budget and closing to within 0.8pp of the never-oracle.

REPRODUCIBILITY STATEMENT

All quantitative claims are anchored to machine-generated Python scripts and `*_out.json` outputs. A historical author-side checker reported 154 matched numerical claims before the final theory-scope revision; the current candidate package contains 40 scripts and 35 frozen outputs, but does not yet contain that checker. Therefore the old count is not a final-package verification: before submission, the authors must rebuild and independently run a claim-to-artifact map against this exact manuscript. All reported runs use six locked seeds and pre-registered endpoints. The candidate scripts, outputs, limitations, compile chain, and the remaining packaging gate are documented in `REPRO_README.md`; see Appendix C for the compute and configuration trail.

ETHICS STATEMENT

This work is theoretical and methodological; it involves no human subjects, no personal or sensitive data, and no applications with foreseeable negative societal impact. The only external dataset is CIFAR-10N (Wei et al., 2022), a public, CC-BY-licensed relabeling of CIFAR-10. Our results concern when training data is absorbed or erased; they carry no privacy, fairness, or dual-use risks beyond those of standard supervised learning.

AI USE STATEMENT

This paper was produced within a multi-agent research system consisting of eight worker agents (A1–A8, powered by large language models), a manager agent, and an external monitor acting in the human-supervision role. Worker execution used Claude Code-compatible harnesses; the current manager and external monitor use Codex. The system operated under human supervision. For this lane (A2), the division of labor was:

- **Research direction, theory, experiments, and drafting.** Agent A2 selected the research question (learning-rate-phase-dependent data value), developed the initial theorem statements, designed and ran all synthetic and real-image experiments, wrote all analysis scripts, and drafted the full manuscript text, figures, and appendices.
- **Independent audit.** Agent A3 performed the Gate-B proof audit of T2/T3 (r787); agent A6 performed a Gate-B pre-audit (r786); near-neighbor citation cross-checking was shared with A6. Audit findings and their resolutions are archived in the lane’s reproducibility trail.
- **External theory-scope repair.** On 2026-08-10 the external monitor found that the current manuscript still overextended the fixed-quadratic, directional-contraction, curvature, and random-reshuffling arguments. An attempted result-independent A2 repair timed out before any write. The monitor then rewrote T1–T3 and C1 to the narrower statements now printed, supplied the detailed proofs, recompiled the PDF, and documented the repair. These revised statements have not yet received an organizationally independent final mathematical sign-off.
- **Human oversight.** Human supervisors defined the task specification, resource boundaries, and gate criteria; adjudicated scope disputes (e.g., directive dc9fce09a81b); and issued the result-driven verdict converting the Gate-C main result to a reversibility-boundary framing (directive ac2720d7313c).
- **Manager coordination.** The manager agent assigned audits, authorized the migration of this draft to the run-level paper directory, and owns final submission decisions.

Quantitative claims are intended to be anchored to machine-generated Python scripts and JSON outputs listed in Appendix C. A historical author-side consistency script reported 154 matched claims before the final theory-scope revision, but that checker is absent from the current candidate package and has not been rerun against these exact manuscript bytes. The final authors must rebuild or recover, inspect, and rerun a claim-to-artifact check before submission. Automated checking does not replace human verification of citations, theorem assumptions, experimental provenance, or interpretation. The human authors and supervisor(s) take full responsibility for the content of this submission, including any errors.

REFERENCES

- Eric Arazo, Diego Ortego, Paul Albert, Noel O’Connor, and Kevin McGuinness. Unsupervised label noise modeling and loss correction. In *ICML*, 2019.
- Lucas Bourtole, Varun Chandrasekaran, Christopher A Choquette-Choo, Hengrui Jia, Adelin Travers, Baiwu Zhang, David Lie, and Nicolas Papernot. Machine unlearning. In *IEEE S&P*, 2021.
- Bo Han, Quanming Yao, Xingrui Yu, Gang Niu, Miao Xu, Weihua Hu, Ivor Tsang, and Masashi Sugiyama. Co-teaching: Robust training of deep neural networks with extremely noisy labels. In *NeurIPS*, 2018.
- Shengding Hu, Yuge Tu, Xu Han, et al. Minicpm: Unveiling the potential of small language models with scalable training strategies. *arXiv:2404.06395*, 2024.
- Andrew Ilyas, Sung Min Park, Logan Engstrom, Guillaume Leclerc, and Aleksander Madry. Data-models: Understanding predictions with data and data with predictions. In *ICML*, 2022.
- Lu Jiang, Zhengyuan Zhou, Thomas Leung, Li-Jia Li, and Li Fei-Fei. Mentornet: Learning data-driven curriculum for very deep neural networks on corrupted labels. In *ICML*, 2018.
- Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *ICML*, 2017.
- Stephan Mandt, Matthew D Hoffman, and David M Blei. Stochastic gradient descent as approximate bayesian inference. *JMLR*, 18, 2017.
- Sung Min Park, Kristian Georgiev, Andrew Ilyas, Guillaume Leclerc, and Aleksander Madry. Trak: Attributing model behavior at scale. In *ICML*, 2023.
- Xuan Qi, Luxi He, Dan Roth, and Xingyu Fu. Dataprophet: Demystifying supervision data generalization in multimodal LLMs. In *International Conference on Learning Representations (ICLR)*, 2026.
- Samuel L Smith and Quoc V Le. A bayesian perspective on generalization and stochastic gradient descent. In *ICLR*, 2018.
- Samuel L Smith, Benoit Dherin, David G T Barrett, and Soham De. On the origin of implicit regularization in stochastic gradient descent. In *ICLR*, 2021.
- Mariya Toneva, Alessandro Sordoni, Remi Tachet des Combes, Adam Trischler, Yoshua Bengio, and Geoffrey J Gordon. An empirical study of example forgetting during deep neural network learning. In *ICLR*, 2019.
- Jiaheng Wei, Zhaowei Zhu, Hao Cheng, Tongliang Liu, Gang Niu, and Yang Liu. Learning with noisy labels revisited: A study using real-world human annotations. In *ICLR*, 2022.

A NEAR-NEIGHBOR AUDIT

Fifteen closest works were checked first-hand (arXiv abstract pages and, where flagged, full PDFs; audit trail NEAR-NEIGHBOR-AUDIT.R774.md with r775/r777 corrections). Each row records whether the work covers our object (LR-phase exposure–timing value decomposition + absorption/erasure non-isomorphism) and the delta. None does.

Work (first-hand source)	Their object	Our delta
Xie et al. 2025 (arXiv:2505.22509)	differentiable stopping time for hyperparameter optimization	data-block 0/1 exposure windows, not hyperparameters; disjoint estimand
DataProphet (arXiv:2603.19688, local PDF)	training-free static metric for dataset-level transfer	we model LR-phase training dynamics; instructionally disjoint
TRAK (Park et al., 2023)	endpoint linearized attribution	we show endpoint attribution is ill-posed under WSD (sign flip) and give the path-functional replacement
TracIn (Pruthi et al. 2020, arXiv:2002.08484)	checkpoint aggregation as a compute mechanism	we study <i>when</i> exposure happens and its saturation law
Forgetting (Toneva et al., 2019)	forgetting events for dataset pruning	exposure saturation law + free-erasure law tied to LR phase
SISA (Bourtole et al., 2021)	sharded retraining for exact unlearning	we give the condition under which the LR tail erases for free
Carlini et al. 2020 (arXiv:2012.07805)	memorization extraction vs. scale	we formalize high-LR-phase memory acquisition via exposure
Datamodels (Ilyas et al., 2022)	endpoint linear surrogates	same delta as TRAK
Smith & Le 2018 (Smith & Le, 2018)	noise scale $g \approx \varepsilon N/B$ and generalization	per-example erasure and phase-dependent value, not noise-scale Bayesian evidence
Mandt et al. 2017 (Mandt et al., 2017)	constant-step SGD as approximate posterior sampling	“when seen” as object; absorption/erasure non-isomorphism
Smith et al. 2021 (Smith et al., 2021)	shuffled-minibatch implicit regularization (generalization side)	within-epoch cyclic coverage causing <i>phase-dependent</i> absorption/erasure (data-value side)
Gradient Starvation (Pezeshki et al. 2021, arXiv:2011.09468)	feature-side learning asymmetry under GD	block-level exposure-window value decomposition
Safran & Shamir 2021 (arXiv:1908.00045)	lower bounds for SGD with random reshuffling (single vs. repeated shuffle; optimization error)	they ask <i>whether</i> shuffling helps convergence; we ask <i>when</i> exposure is absorbed/erased by the LR phase and give a contraction-budget law for the crossing
Nagaraj, Jain & Netrapalli 2019 (arXiv:1903.01463)	sharper without-replacement rates $O(1/K^2)$ for general smooth convex functions	this literature motivates the random-reshuffling comparison; T3 does not derive a without-replacement mean recursion, and our object is phase-dependent block-level value
Mishchenko, Khaled & Richtárik 2020 (arXiv:2006.05988)	“vast improvements” for RR analysis (variance reduction)	they show <i>when RR helps convergence</i> ; we show the LR-phase <i>timing</i> of a block decides its damage, with erasure free in the decay tail
WSD/MiniCPM (Hu et al., 2024)	decay-stage loss drop explained via optimization geometry; their §4.4 explicitly leaves the mechanism open	we identify the decay phase as a <i>data eraser</i> with mechanism (within-epoch cyclic coverage + curvature lower bound) and formalize it

Highest collision risk: Xie et al. (same data-value $\times$ dynamics intersection)—r774 initially mis-described it as a smooth data-weight derivative; first-hand re-check (r775) corrected this: it differentiates a stopping time w.r.t. *algorithm* hyperparameters, disjoint from our data-block exposure functional. The SGD-noise lineage (Smith & Le, 2018; Mandt et al., 2017; Smith et al., 2021) was re-examined after pilot12 falsified our own noise attribution; none of it occupies phase-dependent absorption/erasure. Verdict per the 2026-08-07 protocol: no four-dimensional (problem / mechanism / method / conclusion) isomorphic coverage; similarity alone would not close the line given the explicit deltas above.

B PROOFS

B.1 T1 (THEOREM 1)

Proof. Write $e_t = w_t - w^*$. For the fixed quadratic in T1, $\nabla Q(w_t) = H e_t$, hence $e_{t+1} = (I - \eta_t H) e_t$. Iteration gives the exact matrix product. Because H is fixed and symmetric, it has an orthonormal eigenbasis; the j th coordinate is multiplied by $P_j = \prod_t (1 - \eta_t \lambda_j)$.

For $0 \leq x \leq q < 1$, the Taylor series gives

$$0 \leq -\log(1 - x) - x = \sum_{k=2}^{\infty} \frac{x^k}{k} \leq \frac{x^2}{2(1 - q)}. \quad (7)$$

Set $x = \eta_t \lambda_j$, sum over t , and use $q = q_j = \max_t \eta_t \lambda_j$ to obtain Equation 2. Thus two schedules with the same E have the same leading log contraction, but their exact finite-step products can differ through $\sum_t \eta_t^2$ and higher moments.

If one macro-step of size η is replaced by K substeps of size η/K , then total exposure is still η and $(1 - \eta\lambda/K)^K \rightarrow e^{-\eta\lambda}$; Equation 2 controls the discretization. By contrast, K repetitions of the original step have exposure $K\eta$ and exact factor $(1 - \eta\lambda)^K$, so there is no finite-budget K -invariance. All stable products tend to zero as their cumulative exposure diverges, which explains common-attractor saturation without equating the finite trajectories. $\square$

Scope and diagnostics (T1). The theorem is deliberately fixed-quadratic. In the logistic cell, diag9 reports that a frozen-Hessian calculation matches the observed cross-shape spread to 10^{-5} in one starved regime, and the saturated endpoints match across all tested schedules. Those measurements support approximation quality on the realized paths; they do not upgrade the theorem to arbitrary strongly convex objectives with noncommuting path Hessians.

B.2 T2 (THEOREM 1) AND LEMMA 2

Assumptions. The full-norm theorem requires only the stated segment-wise Hessian bounds and $\eta_t \leq 1/L$. Ridge supplies the global lower bound $\mu = \lambda_r$ in the logistic cell. The faster coordinate statement is separate: it additionally assumes that e_0 is a common eigenvector of every segment-average tail Hessian. The measured damage-direction ratio and curvature are diagnostics for this approximation, not substitutes for the invariant-subspace assumption.

Proof of Lemma 2. The exact directional curvature is $e_0^\top H_A(w) e_0 = \lambda_r + n^{-1} \sum_i q(z_i) x_{i,0}^2$. Because q is even and decreases with $|z|$, the definition $M(w) = \max_i |z_i|$ gives $q(z_i) \geq q(M(w))$ for every sample. Factoring out this common lower bound yields Equation 4. This argument is deterministic and uses the realized $M(w)$ and s_0^2 ; it needs neither a Gaussian norm event nor a claim that a mean logit controls every sample. Direct Hessian-vector diagnostics at five displacement levels and two seeds report positive data curvature on all 10 measured points. The empirical curvature grows from 0.036 to 0.51 along the path; this supports a fast direction on those paths but does not establish common-eigenvector dynamics. $\square$

Proof of Theorem 1. Let $e_t = w_t - w_A^*$. The fundamental theorem of calculus gives $\nabla L_A(w_t) - \nabla L_A(w_A^*) = \bar{H}_t e_t$, where $\bar{H}_t = \int_0^1 \nabla^2 L_A(w_A^* + \tau e_t) d\tau$ and $\mu I \preceq \bar{H}_t \preceq LI$. Thus $e_{t+1} = (I - \eta_t \bar{H}_t) e_t$. For $0 \leq \eta_t \leq 1/L$, every eigenvalue of this symmetric operator lies in $[0, 1 - \eta_t \mu]$, so $\|e_{t+1}\|_2 \leq (1 - \eta_t \mu) \|e_t\|_2$. Iteration and $1 - x \leq e^{-x}$ prove Equation 3.

If $\bar{H}_t e_0 = \lambda_t e_0$ for every tail step, projecting the same recursion onto e_0 gives the exact scalar product and the conditional faster bound with $\lambda_t \geq \lambda_0$. Without invariance, off-diagonal Hessian terms mix other coordinates into e_0 and a Rayleigh lower bound alone cannot justify this projection.

Empirically, diag5 finds $\|w(s) - w_s^{*(D)}\|$ below displayed precision in the measured on-phases, and pilot 9 measures return in parameter space (residual ratio 0.0685), loss (1.017 recovery), and accuracy (1.001). These observations validate the realized return; the theorem’s sufficient condition still depends on the actual E_{tail} and radius R . $\square$

Honest quantitative boundary (pre-registered failures disclosed). The *linearized product law* of step (2) is not pointwise tight: pilot 11’s two pre-registered quantitative criteria both **failed** (V1: predicted/measured residual ratio 0.325 vs bound 0.5; V3: 1.686 vs bound 2). Along the measured trajectory ($t = 120 \rightarrow 230$, 110 steps), a path-averaged linearization *over*-predicts the log shrinkage by $\approx 2.1 \times$ (predicted $\sum \eta_t \lambda_t \approx 1.67$ vs measured $|\Delta \ln D| = 0.785$), while a frozen near-field rate $\lambda_{e_0}(w^*) = 0.037$ under-predicts it (0.328): the true contraction is sandwiched between the two, because the curvature decays along the path as the iterate approaches w_A^* ($\lambda_t : 0.512 \rightarrow 0.094$ — self deceleration). This is the same depth dependence that T3’s $\bar{\lambda}_{\text{path}}$ averages over, and it is why the fast directional rate is reported as an empirical path diagnostic unless the common-invariant-direction premise is verified; the rigorous general rate is μ . The failed V1/V3 are themselves evidence for the sandwich picture and are recorded in the pre-registration verdict file unchanged.

Open items (T2). (i) Proving a faster multidimensional rate requires a common invariant subspace or a quantified coupling remainder; numerically the damage direction satisfies $u \approx e_0$ (projection -0.92), which is not itself a proof. (ii) Non-convex erasure timing is conditional: MLP+SGD retains free erasure (recovery ≥ 0.78) but MLP+GD at a very late drop does not (recovery 0.150 with > 5 cleanup steps needed) — stated as a limitation.

B.3 T3 (THEOREM 2)

Proof. Put $a_t = 1 - \eta_t \lambda$. Because u_t is $\mathcal{F}_t$ -measurable, η_t is deterministic, and $\mathbb{E}[\xi_t | \mathcal{F}_t] = 0$,

$$\mathbb{E}[u_{t+1} | \mathcal{F}_t] = a_t u_t + \eta_t \mathbb{E}[\xi_t | \mathcal{F}_t] = a_t u_t. \quad (8)$$

Taking total expectations gives $\mathbb{E}[u_{t+1}] = a_t \mathbb{E}[u_t]$. Repeated substitution from s to $T - 1$ therefore yields the exact identity

$$\mathbb{E}[u_T] = \mathbb{E}[u_s] \prod_{t=s}^{T-1} a_t = \mathbb{E}[u_s] \prod_{t=s}^{T-1} (1 - \eta_t \lambda). \quad (9)$$

Every factor is positive by assumption. Applying $1 - x \leq e^{-x}$ factor by factor gives

$$\prod_{t=s}^{T-1} (1 - \eta_t \lambda) \leq \exp\left(-\lambda \sum_{t=s}^{T-1} \eta_t\right) = e^{-\lambda B}, \quad (10)$$

which proves Equation 6. For constant η , there are $N_{\text{win}} = T - s$ identical factors, so the exact factor is $(1 - \eta \lambda)^{N_{\text{win}}}$. If $\mathbb{E}[u_s] \neq 0$, the normalized mean crossing $1 - \mathbb{E}[u_T]/\mathbb{E}[u_s]$ is exactly one minus this product.

This proof is deliberately confined to Equation 5. It neither proves that without-replacement logistic SGD has conditionally zero-mean additive noise about a fixed scalar contraction nor controls a non-linear accuracy functional. The GD/SGD and accuracy comparisons below test those two transfer steps empirically. $\square$

Parameter-free verification and its failure (the depth-dependent boundary). A self-fit scalar $\hat{\lambda} = 0.262$ reproduces all four pilot-15 arms but has only one effective degree of freedom (the two $B = 24$ arms saturate for any $\lambda \gtrsim 0.2$), so we do not count it as evidence. The meaningful check is parameter-free: $\bar{\lambda}_{\text{path}} = 0.2023$ from a trapezoidal integral of the pilot-11 curvature sweep $\int_0^1 \lambda(\delta) d\delta$, never fit on pilot 15. It predicts the two $B = 2.4$ arms with residuals $-0.078/-0.085$ and the $B = 24$ arms within 0.007 (verify_t3_ou_out.json, cond_iv passed), and measured equal-budget arms across optimizers match ($|0.4696 - 0.4629| = 0.0067$ GD vs SGD at $B = 2.4$; 0.0002 at $B = 24$). Two additional intermediate-budget GD arms (pilot 17 budget-mid, pre-registered to straddle $B_{\text{crit}} = 1/\bar{\lambda}_{\text{path}} \approx 4.9$): coverage is strictly monotone ($B = 1.2 : 0.249 < 2.4 : 0.470 < 6.0 : 0.864 < 24 : 0.999$) — the ordinal law holds across the seam — but the scalar prediction *under*-predicts the deep-crossing arm ($B = 6$: observed 0.864 vs predicted 0.703, residual -0.161 , marginally outside the pre-registered 0.15 band). This is a boundary of the scalar empirical approximation, not a failure of the affine theorem: the measured effective curvature *grows* with crossing depth ($\lambda(\delta) = \frac{\delta}{1+\delta} \Lambda$, pilot 11), so a shallow-path average necessarily under-rates deep crossings, and the self-fit $\hat{\lambda} = 0.262$ sits between the near-field 0.037

and the far-field saturation 0.51, as a path average must. We therefore claim the scalar law as a robust *ordinal/switch* law (monotonicity across B_{crit} , equal-budget equal-coverage, the $K=1$ anchor), with pointwise accuracy requiring a depth-dependent $\lambda(B)$ — stated as the paper’s quantitative boundary, pre-registered verdicts unchanged.

Empirical seam with T1. In the measured arms, crossing saturates near $\bar{\lambda}_{\text{path}}B \gtrsim 1$, with $B_{\text{crit}} \sim 1/\bar{\lambda}_{\text{path}} \approx 4.9$, and the saturated endpoints become insensitive to the tested schedules. T1 supplies an exact fixed-quadratic product explanation for saturation; T3 supplies an exact affine scalar product. Their shared product shape motivates the empirical seam but does not identify the nonlinear logistic dynamics with either theorem. In particular, $\lambda_{\text{eff}}B \lesssim 1$ is a diagnostic regime in this experiment, not a universal failure certificate for endpoint attribution methods.

C REPRODUCIBILITY DETAILS

Synthetic cell (CPU, seconds per run). The candidate package includes the available `pilot*`, `diag*`, and `verify*.py` scripts and their frozen JSON outputs. All reported experiments use six locked seeds; the author-side audit trail records the endpoint definitions used before each run, but the final submission package must still include and hash the corresponding preregistration records. **Real-image arms (one GPU).** `pilot17/18/19`, `diag20`, `eqra_salvage.py`, `eqra_loss_salvage.py`, `eqra_loss_p3_precision.py`: SmallCNN with WSD (240 epochs, split 120, $\eta_{\text{hi}}=0.05$, $\eta_{\text{lo}}=0.005$, bs 256, momentum 0), MNIST and CIFAR-10 with CIFAR-10N human labels (Wei et al., 2022); 6 seeds, paired arm differences; ~ 10 –25 min per full 6-seed sweep on one H100 (60–70 min for the 60k-example MNIST arms `pilot17/18`). **Verdict documents.** The author workspace contains `GATEC_VERDICT_R785.md`, `DIAG20_VERDICT.md`, `EQRA_COS_VERDICT.md`, `EQRA_LOSS_VERDICT.md`, and `EQRA_LOSS_P3_VERDICT.md`. They are not in the current candidate package and must be independently checked and packaged before submission; this manuscript does not treat their absence as a pass. **Audit trail.** A3’s earlier Gate-B audit (`AUDIT_A2_GATEB_T23_R311.md`) checked the then-current quantitative artifacts. A later external scope audit on 2026-08-10 found theorem-level overreach in the general schedule, directional-contraction, curvature, and random-reshuffling statements. The present manuscript therefore narrows T1 to a fixed quadratic, T2 to full-norm strong-convex return (with a separate invariant-subspace corollary), C1 to a finite-sample max-logit bound, and T3 to the stated affine scalar recursion. These revisions await an independent final mathematical re-audit; the prior Gate-B report is not a sign-off on the revised statements. Raw JSON outputs for every run are alongside the scripts (`*_out.json`).

C.1 DISAGREEMENT-ONLY ROUTER SMOKE (PRE-REGISTERED, RED)

env	never	always	fixed-drop	always-EQRA	dg-EQRA
N (CIFAR-10N)	0.6661	0.6339	0.6651	0.6500	0.6500
H (harmless MNIST block)	0.9891	0.9895	0.9897	0.9649	0.9895
macro	0.8276	0.8117	0.8274	0.8075	0.8197

Table 3: Pre-registered disagreement-gated router smoke (3 fresh seeds, 30/30 arms, medians). The gate uses only repeated-label disagreement computed before training: $\hat{d}=0.396$ on N (route: EQRA, $q=0.10$), $\hat{d}=0.000$ on H (route: always). Frozen GREEN rule: $M \geq M_{\text{base}}+0.005$, $R \leq 0.005$, $N \geq \text{fixed-drop}+0.005$, $H \geq \text{always}-0.005$. Observed regret $R=0.0076$ (per-env loss N 0.0151, H 0.0002) $\Rightarrow$ RED: repeated-label disagreement alone did not safely route the method in this regime (no universal impossibility claimed).

A pre-registered router smoke test (Table 3) asked whether the cheapest available signal — repeated-label disagreement among the three CIFAR-10N human annotations, thresholded before training without access to clean labels, validation, or test outcomes — can select per environment between EQRA-loss quarantine and the no-quarantine default. The gate estimated the harmful fraction as $\hat{d} = 0.396$ on CIFAR-10N (matched quarantine fraction $q = 0.10$) and $\hat{d} = 0.000$ on the harmless MNIST block, routing the former to EQRA($q=0.10$) and the latter to always-on. Under the

frozen decision rule the router is RED: macro accuracy 0.8197 versus the pre-registered fixed-drop comparator 0.8274 (regret $0.0076 > 0.005$), the entire loss concentrated on the noisy environment (-0.0151 versus fixed-drop) while the harmless environment is routed correctly (-0.0002). Repeated-label disagreement therefore conflates annotator ambiguity with harmful mislabeling and overestimates the harmful fraction: disagreement alone did not safely route the current method in this regime. We make no universal impossibility claim — a calibrated estimator (e.g. a per-sample-loss mixture) may still route correctly; the result closes only this disagreement-only gate instance, not the routing question itself.